

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – FLOWCHART-TO-CODE CONVERSION

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 2 – Flowchart-To-Code Conversion
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	PRENA	Reg. No.:	192524056
Section / Batch:	B TECH AI AND DS	Date:	04/09/2026

Code of Conduct

I PRENA (192524056) (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: PRENA M

Instructions

- This test contains 5 Flowchart-to-code conversion questions. Total: 100 Marks.
- When a student answer using Flowchart-to-code conversion should evaluate based on the ability to design clear, logical, and efficient code solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

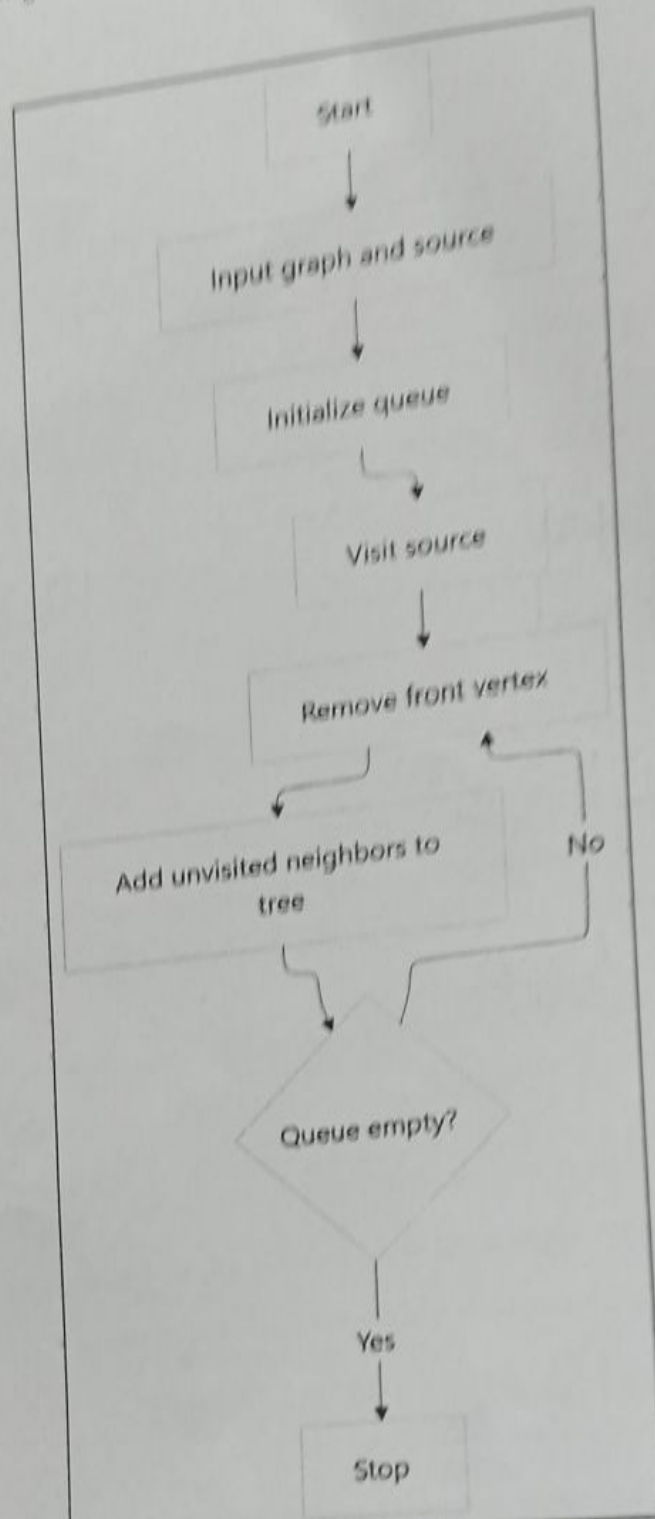
Section / Topic	Focus	Q Nos	Marks
Graph Traversal	BFS	1	20
Shortest Path Algorithm	Dijkstra's Algorithm	2	20
Shortest Path Algorithm	Unweighted Graph	3	20
Graph Traversal	BFS	4	20
Minimum Spanning Tree	Prim's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A - Pseudocode Writing Task

(20 Marks)

1. Convert the flowchart for generating a BFS tree into code.

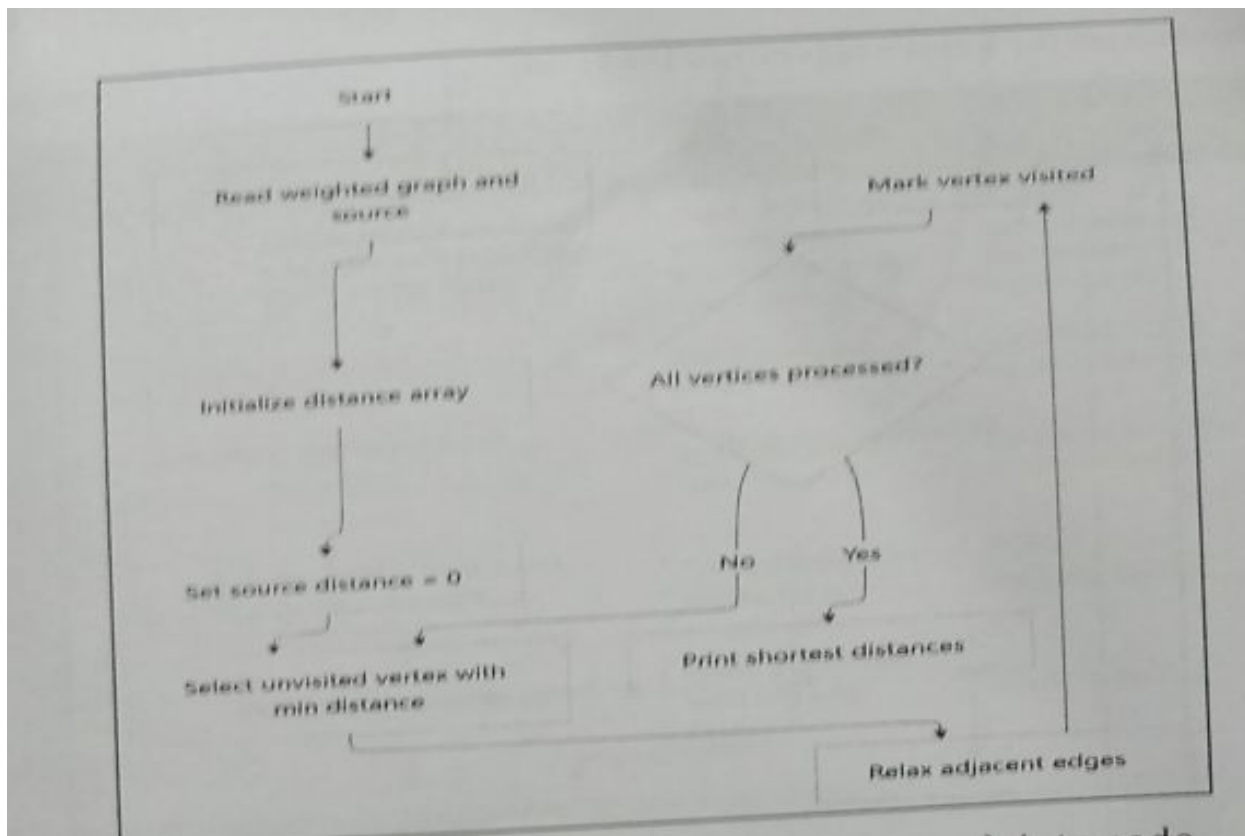
Flowchart Diagram:



2. Convert the flowchart for Dijkstra's shortest path algorithm into code.

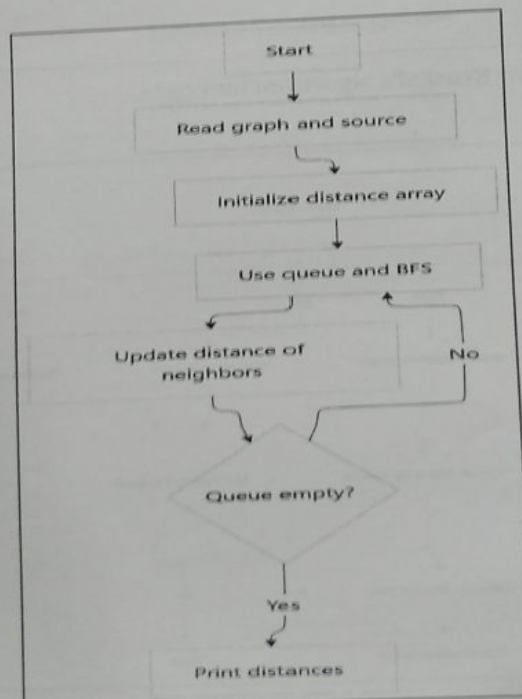
(20 Marks)

Flowchart Diagram:



3. Convert the flowchart for shortest path in an unweighted graph into code. (20 Marks)

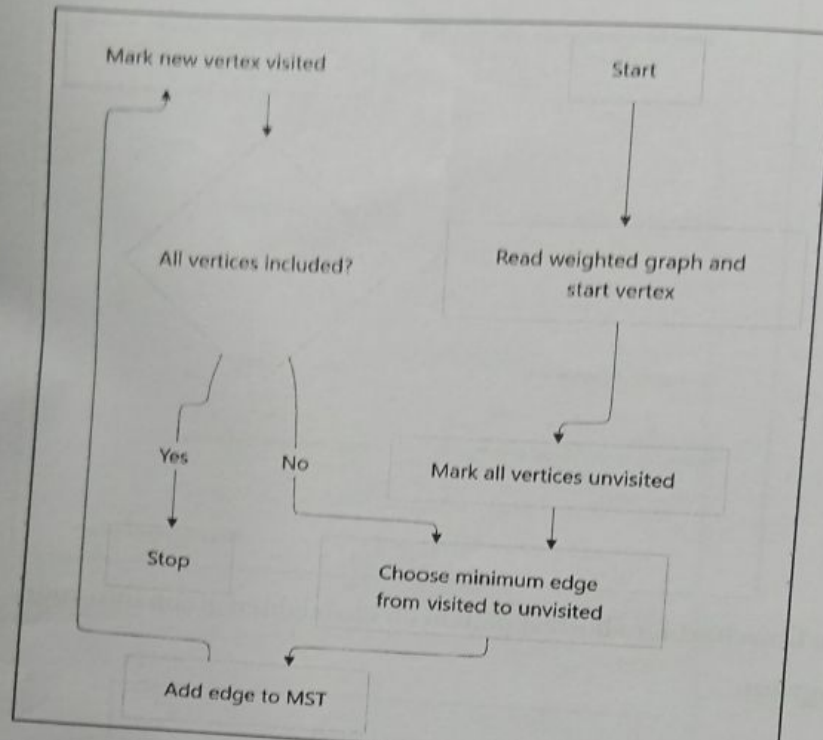
Flowchart Diagram:



4. Convert the flowchart for Prim's algorithm into code.

(20 Marks)

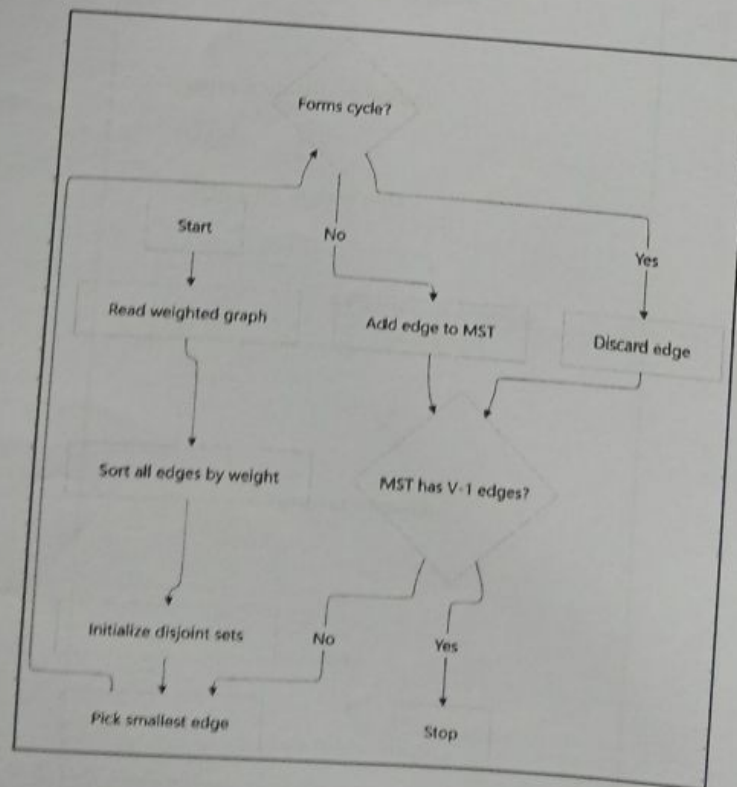
Flowchart Diagram:



5. Convert the flowchart for Kruskal's algorithm into code.

(20 Marks)

Flowchart Diagram:



Rubrics for Calculation

Criteria	Weightage	Level 4- Exemplary	Level 3- Proficient	Level 2- Developing	Level 1- Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Minimal understanding; misses many requirements
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented; follows standards	Readable with some structure	Limited readability; poor structure	Unreadable and disorganized
Complexity Analysis	20	Accurate time/space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty in-charge	Course Coordinator	Program Director
Signature: <u>[Signature]</u>	Signature: _____	Signature: _____
Date: <u>01/09/26</u>	Date: _____	Date: _____